Research Excellence in EEE:

From Methodology to Thesis and Journal Publication

Presenter:
Dr. Abdul Khaleque
Professor, EEE,
&
Director
Research & Extension
RUET

Outline

- Research Idea Development and Research Gap Identification
- Research Methodology and Research Design
- Standard Thesis Writing Guidelines
- Standard Thesis Presentation and Defense Guidelines
- From Thesis to Journal Paper: Standard Journal Writing Guidelines
- Research Excellence, Impact, and Publication Strategy

What is research?

- The creation of new knowledge

  and/or

- The use of existing knowledge in a new and creative way so as to generate new concepts, methodologies and understandings.

Research is a systematic process of investigating a specific topic or problem to discover new facts, gain deeper understanding, or develop new knowledge.

So, select your research area of interest first based on the current need

Research Idea Development

It is systematic process of transforming a broad area of interest into a specific, original, relevant, and researchable problem.

- **Research Interest**
- **Broad Topic**
- **Literature Review**
- **Problem Identification**
- **Research Gap**
- **Research Question**
- **Objectives**
- **Methodology**
- **Research Title**

Basic Steps for Developing a Research Idea

1. Selecting a broad research area

Choose an area related to your academic background, expertise, or current scientific/technological challenges.

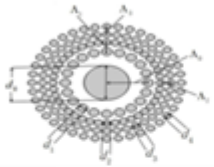
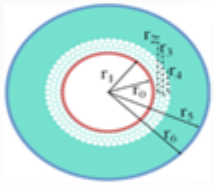
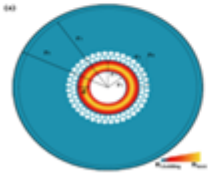
Example:

1. Optical Fiber Communication
2. Perovskite Solar Cell
3. Metamaterial Absorber for Solar Cell Application
4. Plasmonic Solar Cell
5. Antiresonant Optical Fiber
6. Machine Learning based Color Filter
7. ……

2. Exploring recent literature

Review recent journal papers, conference proceedings, review articles, theses, and research reports.

Literature Review:

Sl No.	Year [Ref.]	Fiber Structure	Achievements	Limitations
01	2016 [17]		1. 26 OAM modes 2. Effective refractive index difference above 10^{-4} 3. Low CL (10^{-9} dB/m)	1. Less number of OAM modes 2. High dispersion and nonlinearity 3. Mode purity and quality are not analyzed
02	2021 [18]		1. 180 OAM modes 2. Large refractive index difference above 2×10^{-3} 3. Mode qualities above 94.7 % 4. Low CL (10^{-11} to 10^{-9} dB/m)	1. Manufacturing challenges 2. Mode purity is not analyzed
03	2022 [19]		1. 14 OAM modes 2. Mode purities above 97.05 % 3. Mode qualities above 98.42 % 4. Low CL (10^{-11} dB/m)	1. Less OAM mode number 2. Smaller bandwidth 3. Complex structure

17] Z. A. Hu et al., "Photonic crystal fiber for supporting 26 orbital angular momentum modes," Optics express, vol. 24, no. 15, pp. 17285–17291, 2016.
18] Q. Ma, A. Luo, and W. Hong, "Numerical study of photonic crystal fiber supporting 180 orbital angular momentum modes with high mode quality and flat dispersion," Journal of Lightwave Technology, vol. 39, no. 9, pp. 2971–2979, 2021.
19] X. Zhang, Q. Ma, W. He, W. Hong, and Z. Deng, "Graded ring-core photonic crystal fiber for the transmission of oam modes with high mode purity and quality in c+ l bands," Optics Communications, vol. 520, p. 128486, 2022.

3. Identify important problems

Look for problems that remain unresolved or require improvement.

4. Analyze existing approaches

Examine the methods, results, advantages, and limitations of previous studies.

5. Identify the research gap

Determine what is

- ✓ missing,
- ✓ insufficient,
- ✓ inconsistent, or
- ✓ unexplored.

6. Develop research questions

Converting the gap into specific questions that can be experimentally, theoretically, computationally, or empirically investigated.

7. Define research objectives

- Establishing clear and measurable objectives to answer the research questions.
- Sentence should be grammatically correct
- Maximum around 3/4 objectives

8. Evaluate feasibility

Consider available

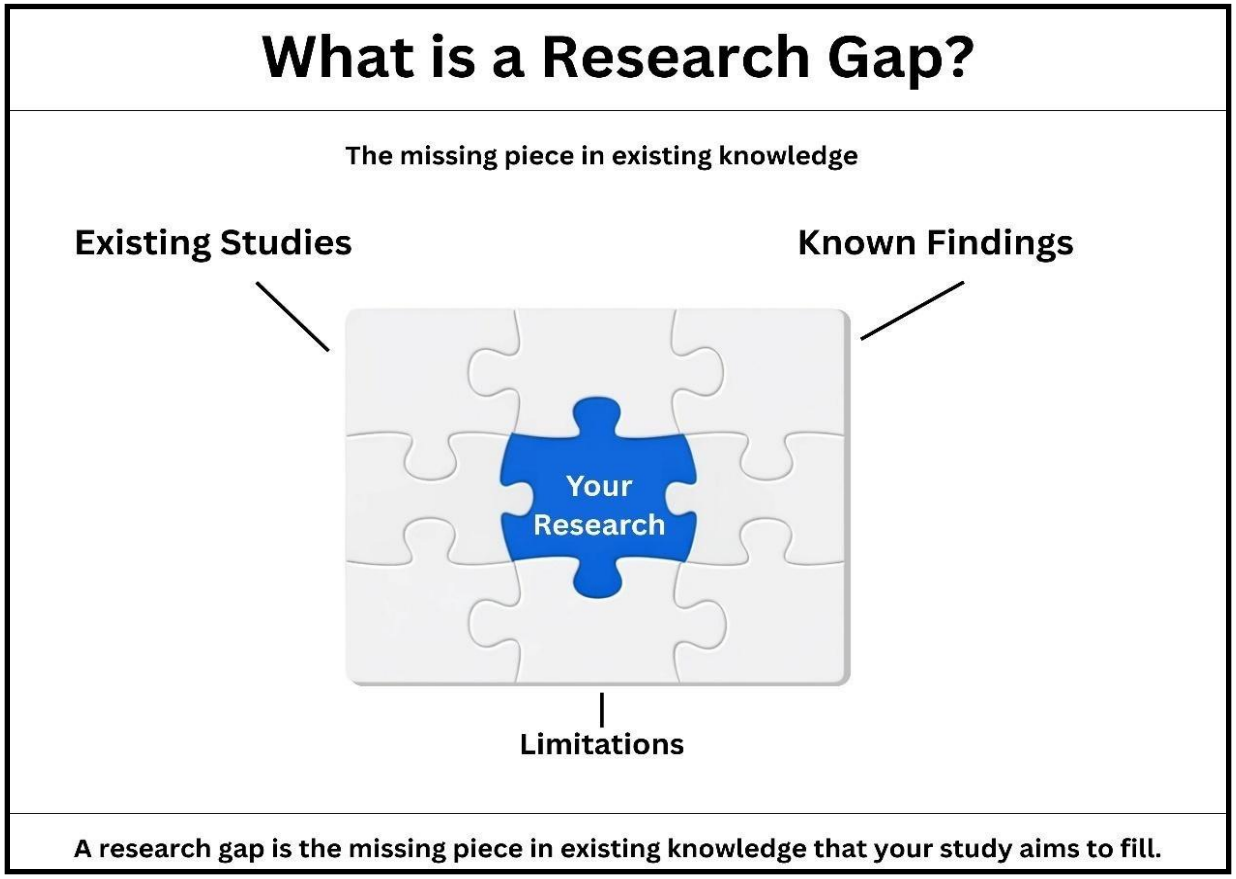
- ✓ equipment,
- ✓ software,
- ✓ data,
- ✓ funding, time, expertise, and other resources.

9. Formulate the research title

Develop a concise title that accurately represents the research problem and proposed contribution.

Research Gap Identification

A **research gap** refers to an area where existing research is **missing, incomplete, inadequate, outdated, contradictory, or insufficiently explored**.



Example

Suppose existing studies have extensively investigated the efficiency of perovskite solar cells.

Research gap :

Despite extensive research on the efficiency and fabrication of perovskite solar cells, limited studies have systematically investigated their long-term stability and degradation mechanisms under the climatic conditions (high-temperature and high-humidity conditions) of Bangladesh.

Research Gap to a research idea:

“Investigation of the Long-Term Stability and Degradation Mechanisms of Perovskite Solar Cells under Bangladesh's Climatic Conditions.”

Major Types of Research Gaps

Research Gap	Description
Knowledge Gap	Important knowledge or information is missing
Methodological Gap	Existing research methods have limitations
Theoretical Gap	Existing theories cannot adequately explain the phenomenon
Empirical Gap	Insufficient experimental or observational evidence exists
Contextual Gap	A topic has not been adequately studied in a particular location or environment
Technological Gap	Existing technology requires improvement or optimization
Contradictory Gap	Previous studies report conflicting findings
Application Gap	Existing knowledge has not been sufficiently applied to practical problems
Temporal Gap	Previous research is outdated and requires investigation using current conditions or technologies

From Research Gap to Research Idea:

Existing Knowledge

↓

Limitations of Existing Research

↓

Research Gap

↓

Research Question

↓

Research Objectives

↓

Proposed Methodology

↓

Expected Contribution

Research Idea Development and Research Gap Identification

Example: Optical Fiber Communication

Research area:
Machine learning in optical fiber communication

Existing research:
Machine-learning techniques have been applied to predict and optimize optical-fiber system performance.

Limitation:
Many existing models require large datasets and may perform poorly when system parameters or operating conditions change.

Research gap:
There is a need for more efficient and generalizable machine-learning models capable of predicting optical-fiber performance under varying system conditions using limited training data.

Research question:
Can a machine-learning model accurately predict optical-fiber performance under varying operating conditions using a relatively small dataset?

Research objective:
To develop and evaluate a computationally efficient machine-learning model for optical-fiber performance prediction.

Example: Optical Fiber Communication

Possible research title:

“Development of a Machine-Learning-Based Model for Performance Prediction of Optical Fiber Communication Systems under Varying Operating Conditions.”

Any Question Please ??

Research Methodology and Research Design

Research Methodology

A systematic approach, principles, techniques, and procedures

used to

collect, analyze, and interpret data

to answer research questions or achieve research objectives.

Major Components of Research Methodology

1. Research problem

- Clearly defines what problem the study intends to investigate.

2. Research questions

- Specifies what the researcher wants to find out.

3. Research objectives

- Defines what the research intends to accomplish.

4. Research approach

- Determines whether the study is qualitative, quantitative, or mixed-methods.

5. Data sources

- Identifies where the required information or data will come from.

6. Data collection methods

- ✓ experiments,
- ✓ surveys, interviews,
- ✓ observations,
- ✓ simulations,
- ✓ measurements,
- ✓ document analysis, etc.

7. Sampling technique

- Determines how participants, samples, datasets, or experimental units will be selected.

8. Data analysis

Specifies the data analysis methods:

- ✓ statistical,
- ✓ computational,
- ✓ mathematical, or
- ✓ qualitative techniques

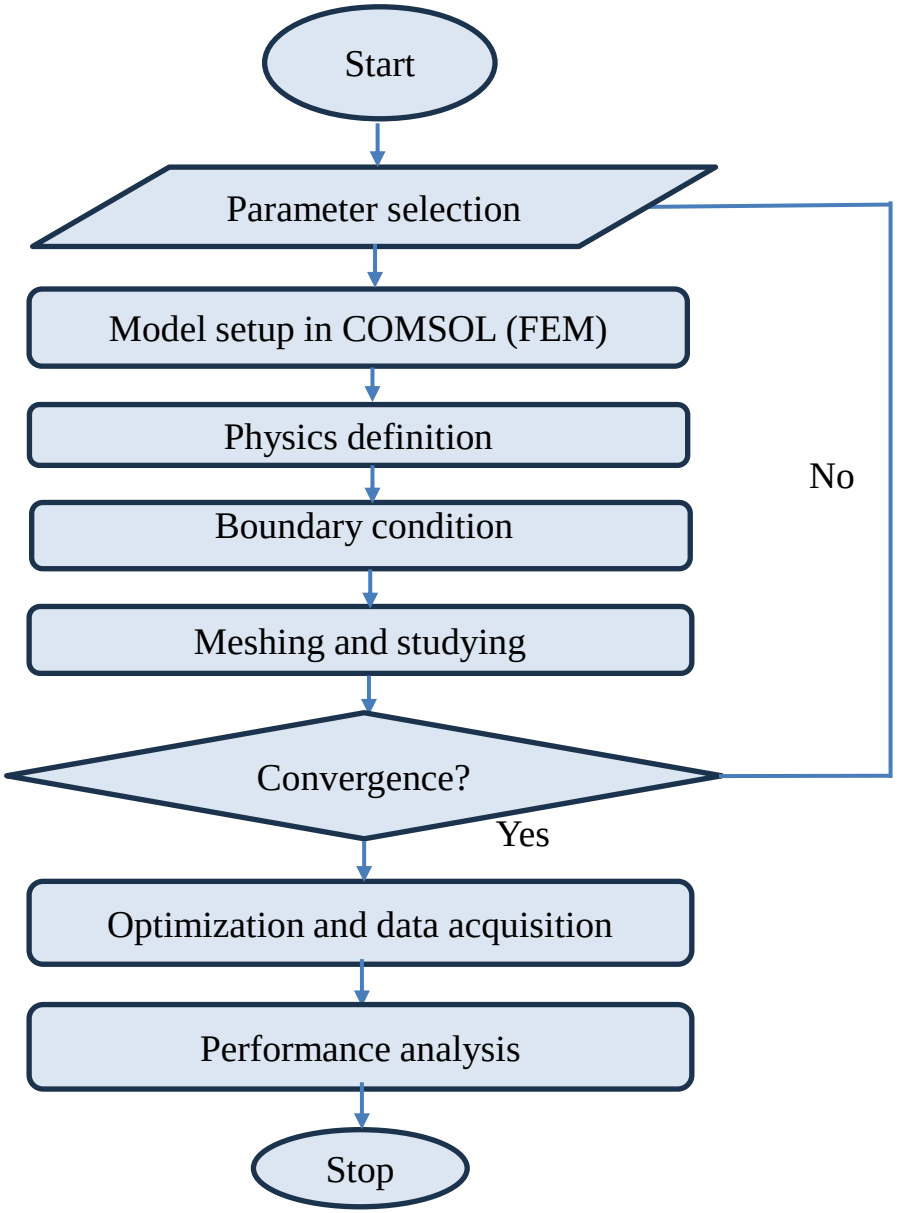
9. Validity and reliability

- Ensures that the research instruments and findings are accurate, consistent, and trustworthy.

10. Ethical considerations

- ✓ informed consent,
- ✓ privacy,
- ✓ confidentiality,
- ✓ data protection, and
- ✓ responsible research practices.

Try to use a flow chart for methodology, if possible



Research Design

Research design is the overall **plan or framework** that specifies how a research study will be structured and conducted.

Research Methodology	Research Design
Explains the overall approach and methods used in research	Provides the structural plan of the study
Explains how and why particular methods are selected	Explains how the study will be organized
Broader concept	More specific component
Includes research approach, data collection, analysis, ethics, etc.	Includes experimental structure, sampling plan, variables, procedures, etc.
Provides justification for methodological choices	Provides the practical framework for implementing them

Research Design : Types

A. Experimental Research Design

Used when the researcher manipulates one or more variables and observes their effects.

Example: Investigating how temperature affects the efficiency of a solar cell.

B. Descriptive Research Design

Used to describe the characteristics, conditions, or behaviors of a particular population or phenomenon.

Example: A survey of renewable-energy adoption among industries.

C. Correlational Research Design

Used to determine whether and how strongly two or more variables are related.

Example: Investigating the relationship between temperature and solar-cell efficiency.

D. Survey Research Design

Uses questionnaires or structured interviews to collect information from a population or sample.

Example: Studying researchers' perceptions of artificial intelligence in engineering education.

E. Case Study Design

Provides an in-depth investigation of a specific organization, system, community, technology, or event.

Example: A case study of solar-energy integration in a particular university.

F. Qualitative Research Design

Used to understand experiences, perceptions, opinions, behaviors, and meanings.

- Interviews
- Focus groups
- Observations
- Document analysis

G. Quantitative Research Design

Uses numerical data, measurements, mathematical models, and statistical analysis.

Examples include:

- Experimental measurements
- Surveys with numerical scales
- Statistical modeling
- Simulation results

H. Mixed-Methods Research Design

Combines qualitative and quantitative approaches.

For example:

Survey → Statistical analysis → Interviews → Thematic analysis → Integrated interpretation

Example of a Complete Research Design

Research Topic : Machine-Learning-Based Performance Prediction of Optical Fiber Communication Systems

Research Objective : To develop a machine-learning model for predicting the performance of an optical-fiber communication system.

Research Design

Step 1: Define system parameters
↓
Step 2: Generate experimental/simulation data
↓
Step 3: Pre-process and normalize data
↓
Step 4: Divide data into training and testing datasets
↓
Step 5: Develop machine-learning models
↓
Step 6: Train the models
↓
Step 7: Test and validate the models
↓
Step 8: Compare prediction performance
↓
Step 9: Statistical error analysis
↓
Step 10: Identify the best-performing model
↓
Step 11: Draw conclusions and recommendations

Any Question Please ??

Standard Thesis Writing Guidelines

A **standard thesis** should present a clear research problem, demonstrate knowledge of existing literature, identify a research gap, describe a rigorous methodology, present and analyze results, and explain the contribution of the research.

1. Standard Thesis Structure

Preliminary Pages

1. **Title Page**
2. **Declaration**
3. **Certificate/Approval Page**
4. **Dedication** *(optional)*
5. **Acknowledgements**
5. **Abstract**
6. **Table of Contents**
7. **List of Figures**
8. **List of Tables**
9. **List of Abbreviations/Symbols**

Writing the Abstract

Background → Problem → Objective → Methodology → Key Results → Conclusion → Contribution / Application

❖ Approximately **200–350 words**, unless the university specifies a different limit.

Avoid:

- Detailed literature review
- Long explanations
- References
- Unnecessary abbreviations
- Information that does not appear in the thesis

Main Chapters

Chapter 1: Introduction

- Background of the study
- Problem statement
- Research motivation
- Research gap
- Research questions
- Research objectives
- Scope of the study
- Significance of the research
- Limitations
- Organization of the thesis

Chapter 2: Literature Review

- Theoretical background
- Previous research
- Current state of the field
- Comparison of existing approaches
- Limitations of previous studies
- Identification of the research gap
- Summary of the literature review

Chapter 3: Research Methodology

- Research approach
- Research design
- Materials/equipment/software
- Data sources
- Sampling, where applicable
- Experimental or computational procedure

- Data collection
- Data analysis techniques
- Validation and reliability
- Ethical considerations, where applicable

Chapter 4: Results and Analysis

Results:

- Presentation of results
- Tables and figures
- Statistical analysis
- Experimental/simulation results
- Comparison with existing methods
- Interpretation of findings

Discussion:

- Interpretation of major findings
- Comparison with previous research
- Theoretical implications
- Practical implications
- Explanation of unexpected results
- Research contribution

Chapter 5: Chapter required for OBE conditions (If any)

- Social and environmental influence
- Addressing Program Outcomes, Complex Engineering Problems and Activities.

Main Chapters

Chapter 6: Conclusions and Recommendations

- Summary of major findings
- Achievement of objectives
- Main contributions
- Conclusions
- Limitations
- Recommendations
- Future research directions

End Matter

- **List of Publications**
- **References**
- **Appendices**

List of Publications: Phase-I, II, III

Journals:

1. **Rosni Sayed**, Abdul Khaleque, Kumary Sumi Rani Shaha, Md. Anowar Kabir, and Rifat Ahmmed Aoni, "Dual step-index ring core antiresonant fiber for broadband transmission of numerous OAM modes with high purity and quality," **Opt. Mater. Express** 15, 379-394 (2025)

2. **Rosni Sayed**, Abdul Khaleque, and Kumary Sumi Rani Shaha, "Low crosstalk holey trench-assisted multi-core fiber supporting over 500 OAM modes," **Opt. Express** 33, 31437-31453 (2025).

3. Kumary Sumi Rani Shaha, Munira Islam, Abdul Khaleque, Azra Sadia Sultana, Mst. Sumaya Akter, **Rosni Sayed**, Extremely low loss and broad bandwidth in an interconnected nested tube antiresonant fiber, **Optics Communications**,Volume 605, 2026, 132863.

4. Sakib Sadman, Abdul Khaleque, Rabiul Islam, and **Rosni Sayed**, "Deep neural network for color prediction from a dielectric metasurface and inverse design of corresponding color filter," **Opt. Mater. Express** 15, 2308-2323 (2025)

Use the citation style required by your institution, such as:

- IEEE
- APA
- Harvard
- Vancouver
- Chicago

Referencing and Plagiarism

All ideas, data, figures, equations, or statements taken from other sources should be appropriately acknowledged.

Avoid plagiarism by:

- Properly paraphrasing sources.
- Citing original sources.
- Using quotation marks for direct quotations.
- Not copying text and merely changing a few words.
- Citing reused or adapted figures and tables where required.

Standard Thesis Writing Principles

A. Clarity

Write clearly and directly. Avoid unnecessarily complicated sentences.

B. Logical Organization

Each chapter should have a logical connection with the previous and following chapters.

C. Academic Language

Use formal, objective, and professional academic language.


"We got very good results."

Use the following:

"The proposed method demonstrated improved performance compared with the existing approach."

D. Evidence-Based Statements

Important claims should be supported by appropriate scholarly references, experimental evidence, mathematical analysis, or reliable data.

E. Consistency

Maintain consistent:

- Terminology
- Symbols
- Units
- Figure numbering
- Table numbering
- Citation style
- Headings
- Formatting
- Spelling

Thesis Formatting Guidelines

Unless your university provides specific requirements, a common standard is:

Element	Typical Guideline
Paper size	A4
Font	Times New Roman
Main text	12 pt
Chapter heading	16–18 pt, bold
Section heading	14 pt, bold
Line spacing	1.5
Alignment	Justified
Paragraph	Consistent first-line indentation
Margins	Usually 1–1.5 inches
Page numbering	Consistent throughout
Figures	Numbered and captioned
Tables	Numbered and titled
Equations	Sequentially numbered

Any Question Please ??

Standard Thesis Presentation and Defense Guidelines

Objectives of the Thesis Defense

The presentation and defense should demonstrate that the researcher:

- Understands the research problem and its significance.
- Has critically reviewed the relevant literature.
- Has identified a valid research gap.
- Has selected an appropriate research methodology.
- Can explain and justify the research design.
- Has obtained and analyzed reliable results.
- Understands the limitations of the research.
- Can clearly explain the original contribution of the study.
- Can respond confidently and logically to questions from the examination committee.

Standard Thesis Presentation Structure

A recommended presentation sequence is:

Slide 1: Title

Include:

- Thesis title
- Researcher's name
- Student ID, if required
- Department
- University/institution
- Supervisor/co-supervisor
- Date

Slide 2: Research Background

Explain:

- The broader research area
- Current situation
- Importance of the topic
- Motivation for conducting the research

Slide 3: Problem Statement

Clearly identify:

- The existing problem
- Why the problem is important
- Consequences of the problem
- The need for investigation

Slide 4: Research Motivation

Explain **why this particular research was undertaken**.

Connect the motivation to:

Existing Problem → Need for Solution → Research Opportunity

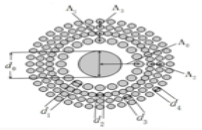
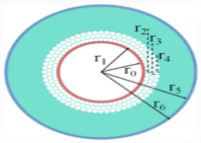
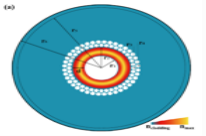
Slide 5: Literature Review

Present only the **most relevant previous studies**.

Focus on:

- (i) Major approaches (ii) Important findings
- (iii) Existing technologies/methods (iv) Limitations of previous research
- (v) Avoid presenting a long list of papers.

Literature Review: Phase-I

Sl No.	Year [Ref.]	Fiber Structure	Achievements	Limitations
01	2016 [17]		1. 26 OAM modes 2. Effective refractive index difference above 10^{-4} 3. Low CL (10^{-9} dB/m)	1. Less number of OAM modes 2. High dispersion and nonlinearity 3. Mode purity and quality are not analyzed
02	2021 [18]		1. 180 OAM modes 2. Large refractive index difference above 2×10^{-3} 3. Mode qualities above 94.7 % 4. Low CL (10^{-11} to 10^{-9} dB/m)	1. Manufacturing challenges 2. Mode purity is not analyzed
03	2022 [19]		1. 14 OAM modes 2. Mode purities above 97.05 % 3. Mode qualities above 98.42 % 4. Low CL (10^{-11} dB/m)	1. Less OAM mode number 2. Smaller bandwidth 3. Complex structure

17] Z. A. Hu et al., "Photonic crystal fiber for supporting 26 orbital angular momentum modes," Optics express, vol. 24, no. 15, pp. 17285–17291, 2016.
18] Q. Ma, A. Luo, and W. Hong, "Numerical study of photonic crystal fiber supporting 180 orbital angular momentum modes with high mode quality and flat dispersion," Journal of Lightwave Technology, vol. 39, no. 9, pp. 2971–2979, 2021.
19] X. Zhang, Q. Ma, W. He, W. Hong, and Z. Deng, "Graded ring-core photonic crystal fiber for the transmission of oam modes with high mode purity and quality in c+ l bands," Optics Communications, vol. 520, p. 128486, 2022.

Slide 6: Research Gap

What is known

↓

What previous studies have done

↓

What is still missing

↓

Research gap addressed by this thesis

Slide 7: Research Questions

- ✓ Present the main research questions clearly and concisely.
- ✓ Each question should be directly related to the research problem and gap.

Slide 8: Research Objectives

- Main objective
- Specific objectives
- ➢ The objectives should correspond directly to the research questions.
- ➢ Maximum 3 / 4 objectives in bulletin format is enough

Slide 9: Research Methodology

- Research approach
- Research design
- Data/material sources
- Experimental or computational methods
- Data collection
- Data analysis

Try to use a **flowchart, block diagram, ……**

Slide 10: Research Framework/Experimental Setup

Whenever possible, use a

- ✓ **flowchart,**
- ✓ **block diagram,**
- ✓ **system architecture, or**
- ✓ **experimental setup**

Example:

Input → Processing/Experiment → Analysis → Validation → Output

Slide 11–13: Results & Discussion

Results

Use:

- Graphs
- Tables
- Images
- Statistical results
- Performance comparisons

Avoid showing too many similar results.

Discussion

- What the results mean
- Why the results occurred
- How they compare with previous studies
- Whether the objectives were achieved
- The significance of the findings

Slide 14: Research Contributions

What is new in this research?

- New methodology
- New model
- Improved algorithm
- New experimental findings
- Improved performance
- New theoretical understanding
- Practical application

Slide 15: Limitations

Be honest about:

- Experimental limitations
- Dataset limitations
- Model assumptions
- Equipment limitations
- Scope limitations
- Generalizability

Acknowledging limitations demonstrates **research maturity**, not weakness.

Slide 16: Conclusions

- ✓ Summarize the most important findings.
- ✓ The conclusion should directly address the research questions and objectives.

Slide 17: Future Work

Identify realistic future research directions.

- Larger datasets
- Improved methodology
- Different operating conditions
- Real-world implementation
- Long-term testing
- Alternative models or techniques

Slide 18: Publications/Research Outputs

If applicable, include the list of:

- Journal papers
- Conference papers
- Patents
- Software
- Datasets
- Research awards

Slide 19: Acknowledgement of the Supports

Acknowledgement

This research work is supported by:

- **Department of Electrical & Electronic Engineering, Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh.**
- **Science and Technology Fellowship Trust (STFT), Ministry of Science and Technology, Bangladesh** under the **PhD Fellowship Program of 2022-2023.**
- **Office of Research & Extension (R&E), RUET** under the **PhD Program at EEE Department of 2022-2023.**

The authors gratefully acknowledge the financial and institutional support.

Slide 20: Thank You / Questions

- ✓ **Thank You**
- ✓ **Questions and Discussion**

Thesis Defense Question Categories

The examination committee may ask questions from several areas.

A. Research Problem

- Why did you select this topic?
- Why is this problem important?
- Who will benefit from the research?
- What motivated the study?

B. Literature Review

- What are the major findings from previous studies?
- Which studies are most relevant to your work?
- What are the limitations of existing approaches?
- How did you identify the research gap?

C. Research Gap

- What exactly is the research gap?
- Why is the gap important?
- How does your research address the gap?
- What evidence supports your claim that the gap exists?

D. Methodology

- Why did you choose this methodology?
- Why did you select this research design?
- Why did you choose this dataset/material/equipment?
- What alternative approaches could have been used?
- How did you validate the methodology?

E. Results

- What is your most important finding?
- Which result is most significant?
- Why did you obtain this result?
- Are the results consistent with previous research?
- How reliable are your findings?

F. Originality

- What is novel about your research?
- What is your main contribution?
- How is your work different from previous studies?

G. Limitations

- What are the major limitations?
- What assumptions did you make?
- How might these limitations affect the findings?

H. Future Research

- What would you do differently in future?
- How can this research be extended?
- Can your approach be applied in real-world situations?

How to Answer Defense Questions

Direct Answer → Evidence → Explanation → Limitation/Future Direction Example

Example:

Question: Why did you select this methodology?

Answer:

- ✓ "This methodology was selected because it is appropriate for addressing the research objectives and allows the required variables to be evaluated under controlled conditions.
- ✓ The comparative results support its suitability for the present study.
- ✓ However, its applicability under conditions outside the scope of this research requires further investigation."

If you do not know the answer:

Do **not** invent an answer.

A professional response is:

- ✓ "That is an important question.
- ✓ It was outside the scope of the present study, so I did not investigate it directly.
- ✓ However, it would be an appropriate direction for future research."

Presentation Time Management

Section	Approximate Share
Background & Problem	10–15%
Literature & Gap	10–15%
Objectives	5%
Methodology	15–20%
Results & Discussion	30–40%
Contribution & Conclusion	10–15%

Common Mistakes to Avoid

Presentation Mistakes

- Reading directly from slides.
- Using too much text.
- Presenting every literature paper.
- Showing unreadable graphs.
- Using excessive animations.
- Spending too much time on background.
- Failing to explain the research gap.
- Presenting results without interpretation.
- Not clearly stating the contribution.

Defense Mistakes

- Becoming defensive when criticized.
- Guessing answers.
- Making unsupported claims.
- Blaming equipment, supervisors, or previous researchers.
- Failing to acknowledge limitations.
- Giving unnecessarily long answers.
- Answering a different question from the one asked.

Any Question Please ??

Thesis to Journal Paper: Standard Journal Writing Guidelines

Converting a thesis into a journal paper requires:

- ✓ **selecting the strongest research contribution**

and

- ✓ **presenting it in a concise, focused, and publication-ready format.**

A journal paper should not simply be a shortened version of the thesis.

Thesis vs. Journal Paper

Thesis	Journal Paper
Comprehensive	Focused
Detailed literature review	Selective literature review
Extensive methodology	Concise but reproducible methodology
Many results may be included	Only key results are emphasized
Multiple objectives may be addressed	Usually centered on a specific contribution
Detailed explanations	Concise scientific argument
Written for degree assessment	Written for the research community

Standard Journal Paper Structure

Title → Abstract → Keywords → Introduction → Literature Review/Related Work → Design & Methodology → Results & Discussion → Conclusion → References

1. Title

The title should be:

- Specific
- Concise
- Informative
- Scientifically accurate
- Relevant to the main contribution

Avoid

"A Study on Some Important Aspects of Solar Cells"

Better way:

"Experimental Investigation of Temperature-Dependent Performance Degradation in Perovskite Solar Cells"

The title should allow readers to understand the **topic, system, method, or contribution** immediately.

2. Abstract

Background → Problem/Gap → Objective → Method → Key Results → Conclusion/Contribution

Recommended characteristics

- Concise
- Self-contained
- Quantitative where appropriate
- No unnecessary background
- Usually no references
- Avoid unexplained abbreviations

A strong abstract should answer:

- **What problem was addressed?**
- **How was it investigated?**
- **What was found?**
- **Why does it matter?**

3. Keywords

Select approximately **4–8 important keywords**, depending on journal requirements.

Keywords should represent:

- Research topic
- Main technology
- Methodology
- Important variables
- Application area

For example:

Keywords: Perovskite solar cell; degradation; stability; temperature; humidity; photovoltaic performance.

4. Introduction

The introduction should establish a logical argument:

Recommended structure

- ✓ **Paragraph 1 – Context**
  Introduce the broad research area.
- ✓ **Paragraph 2 – Current research**
  Discuss important recent developments.
- ✓ **Paragraph 3 – Problem and limitations**
  Explain what remains problematic.
- ✓ **Paragraph 4 – Research gap**
  Clearly identify what is missing.
- ✓ **Final paragraph – Present study**
  State:
  - What this paper investigates
  - How it addresses the gap
  - Main contribution

5. Literature Review / Related Work

A journal paper generally does not require the extremely detailed literature review found in a thesis.

- Select the most relevant studies.
- Prioritize recent and high-quality publications.
- Compare competing approaches.
- Identify limitations.
- Establish the research gap.

Avoid

A sequence such as:

Author A did this. Author B did this. Author C did this. Author D did this.

Prefer synthesis

"Previous studies have demonstrated that X improves performance; however, these approaches generally require Y and have limited applicability under Z conditions."

This demonstrates **critical analysis rather than literature description**.

6. Methodology / Materials and Methods

The methodology should be **concise but sufficiently detailed for scientific evaluation and reproducibility**.

- Research design
- Materials
- Equipment
- Experimental setup
- Dataset
- Software

- Mathematical model
- Algorithms
- Simulation parameters
- Data collection procedure
- Statistical methods
- Evaluation metrics

7. Results & Discussion

Present the results that directly support the research objectives.

Use:

- Figures
- Tables
- Graphs
- Statistical analysis
- Performance metrics
- Comparative analysis

Good practice

Do not simply state:

"The proposed model performed better."

Instead, provide evidence:

"The proposed model achieved an accuracy of X%, compared with Y% for the baseline model."

Discussion

What do these results mean?

1. Major findings.
2. Reasons for the observed results.
3. Comparison with previous studies.
4. Agreement or disagreement with existing knowledge.
5. Theoretical significance.
6. Practical significance.
7. Limitations and possible explanations.

Strong discussion structure

Finding → Explanation → Comparison → Significance

For example:

"The proposed method improved prediction accuracy compared with the baseline model. This improvement may be attributed to… The finding is consistent with previous studies reporting… However, unlike those studies, the present work…"

Results and Discussion: Avoid Repetition

Do not repeat the same information unnecessarily.

Results:

What happened?

Discussion:

Why did it happen, and why does it matter?

This distinction makes the paper more concise and scientifically stronger.

8. Conclusion

- Summarize the major findings.
- Address the research objective.
- State the principal contribution.
- Mention practical or theoretical implications.
- Identify appropriate future work.
- Avoid introducing **new results or arguments** in the conclusion.

A strong conclusion should answer:

What did the study discover, and why is that discovery important?

erica B Research Article

Acknowledgment. The authors would like to sincerely thank the Department of EEE, Rajshahi University of Engineering & Technology, Rajshahi-6204, Bangladesh, for their contribution and support.

Disclosures. The authors certify that there are no competing interests associated with this work.

Data availability. The data supporting the findings of this study are currently not publicly accessible; however, the dataset may be provided by the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

REFERENCES

1. F. Zhang, J. Martin, S. Murai, *et al.*, "Broadband scattering by an alu-

Figures and Tables

Every figure and table should have a clear purpose.

Figures should:

- Be high resolution.
- Have readable labels.
- Use consistent units.
- Have appropriate captions.
- Be referenced in the text.
- Avoid unnecessary decoration.

Tables should:

- Have clear headings.
- Use consistent decimal places.
- Include units where necessary.
- Avoid excessive information.

Do not present the same data unnecessarily in **both a table and a graph**.

References

Use the citation style required by the target journal.

Common styles:

- IEEE
- APA
- Vancouver
- Harvard
- Chicago

Good reference practices

- Use recent and relevant literature.
- Cite original research where appropriate.
- Include foundational studies when necessary.
- Verify every citation.
- Ensure every in-text citation appears in the reference list.
- Ensure every reference in the list is cited in the manuscript.

Selecting a Suitable Journal

Before submission, evaluate:

- Journal scope
- Recent articles in the same research area
- Indexing status
- Peer-review process
- Publication model

- Article processing charges
- Acceptance/rejection policies
- Expected publication timeline
- Author guidelines
- Ethical and research-integrity requirements

Do not select a journal based only on its impact factor.

Common Mistakes in Thesis-to-Journal Conversion

Please Avoid the Followings:

- ✕ Copying thesis chapters directly
- ✕ Excessively long literature review
- ✕ Including every thesis result
- ✕ Weak research-gap statement
- ✕ Unclear novelty
- ✕ Unsupported claims

- ✕ Poor-quality figures
- ✕ Outdated references
- ✕ Inconsistent terminology
- ✕ Ignoring journal formatting requirements
- ✕ Overclaiming the significance of findings
- ✕ Submitting to an inappropriate or questionable journal

Any Question Please ??

Research Excellence, Impact, and Publication Strategy

Research Excellence, Impact, and Publication Strategy

- **Research excellence** is not simply the production of a large number of papers.

It involves conducting

- ✓ **high-quality,**
- ✓ **original,**
- ✓ **rigorous,**
- ✓ **relevant, and**
- ✓ **ethically responsible research**

that contributes meaningfully to knowledge and society.

Key aspects of Research Excellence

Dimension	Key Area
Originality:	Producing new ideas, methods, or knowledge.
Research quality:	Using sound theories, methods, data, and analysis.
Rigor:	Conducting research carefully, systematically, and objectively.
Innovation:	Developing new approaches, technologies, or solutions.
Impact:	Making a meaningful contribution to science, society, industry, or policy.
Publication quality:	Publishing findings in reputable journals or conferences.
Ethics and integrity:	Following ethical standards and maintaining honesty and transparency.
Collaboration:	Working effectively with other researchers and institutions.

What is Research Impact?

Research impact refers to the changes or benefits that result from research beyond simply publishing the findings.

Research Outputs vs. Research Impact

Output:

Published journal article.

Impact:

The research method is adopted by other researchers, improves an industrial process, influences policy, contributes to a new technology, or produces measurable societal benefits.

Publication is an output; impact is the consequence or influence of the research.

Categories of Research Impact:

A. Academic Impact

Measured through:

- Citations
- Use of research by other scholars
- Follow-up studies
- Collaborations
- Conference invitations
- Research networks

B. Technological Impact

Examples:

- New device
- New algorithm
- New software
- Prototype
- Patent
- Improved engineering process

C. Industrial Impact

Examples:

- Technology adoption
- Cost reduction
- Improved efficiency
- Improved reliability
- New products or services

D. Societal Impact

Examples:

- Improved public services
- Environmental benefits
- Improved quality of life

Educational improvements

E. Policy Impact

Examples:

- Government reports
- Technical standards
- Policy recommendations
- Evidence used in decision-making

Publication Strategy

A **publication strategy** is a planned approach for converting research outputs into high-quality scholarly publications.

Instead of asking:
"How many papers can I publish?"

Please ask:

"What research contribution can I communicate, to which audience, and through which appropriate publication?"

Develop a Publication Pipeline:

Research Idea
↓
Literature Review
↓
Research Gap
↓
Research Project
↓
Preliminary Results
↓
Conference Paper / Workshop
↓
Full Journal Manuscript
↓
Peer Review
↓
Revision
↓
Publication
↓
Research Dissemination
↓
Impact & Follow-up Research

Selecting the Right Journal

Journal selection should be based on **fit and quality**, not only journal metrics

Evaluate:

Scope

Does the journal regularly publish research similar to yours?

Quality

Examine the journal's editorial standards and published papers.

Indexing

Check whether it is indexed in databases relevant to your field.

Audience

Will the researchers who need your work actually read this journal?

Publication Model

- Subscription
- Open access
- Hybrid publishing

Publication Cost

Check article processing charges and other fees.

Review Process

Consider the journal's peer-review model and expected timeline.

Author Guidelines

Check:

- Word limit
- Figures
- References
- Manuscript format
- Article type
- Ethical requirements

Quality Over Quantity

A strong publication strategy prioritizes:

High-quality, coherent, and meaningful publications rather than unnecessary fragmentation of one study into many papers.

A better approach is to identify genuinely distinct contributions.

1. **Methodology paper** – new method or model.
2. **Experimental paper** – experimental validation.
3. **Application paper** – real-world implementation.

Publication Ethics

Research excellence requires strong research integrity.
Researchers should avoid:

- Plagiarism
- Fabrication of data
- Falsification of results
- Duplicate publication
- Unethical authorship
- Undisclosed conflicts of interest
- Manipulation of citations
- Simultaneous submission of the same manuscript to multiple journals

Authorship

Authorship should reflect **substantial intellectual or research contributions** according to the relevant journal and institutional standards.

Thanks for your attention.

Any Question Please ??